

AIM : Message Authentication Codes

Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

CLI Version :

Code :

```
import hmac
import hashlib

# User enters the secret key
secret_key = input("Enter Secret Key: ").encode()

# User enters the message
message = input("Enter Message: ").encode()

# Generate MAC
mac = hmac.new(secret_key, message, hashlib.sha256).hexdigest()

print("\nGenerated MAC:", mac)

# ----- Verification -----

print("\n----- Verify Message -----")

received_message = input("Enter Received Message: ").encode()
received_mac = input("Enter Received MAC: ")

# Generate MAC again
new_mac = hmac.new(secret_key, received_message, hashlib.sha256).hexdigest()

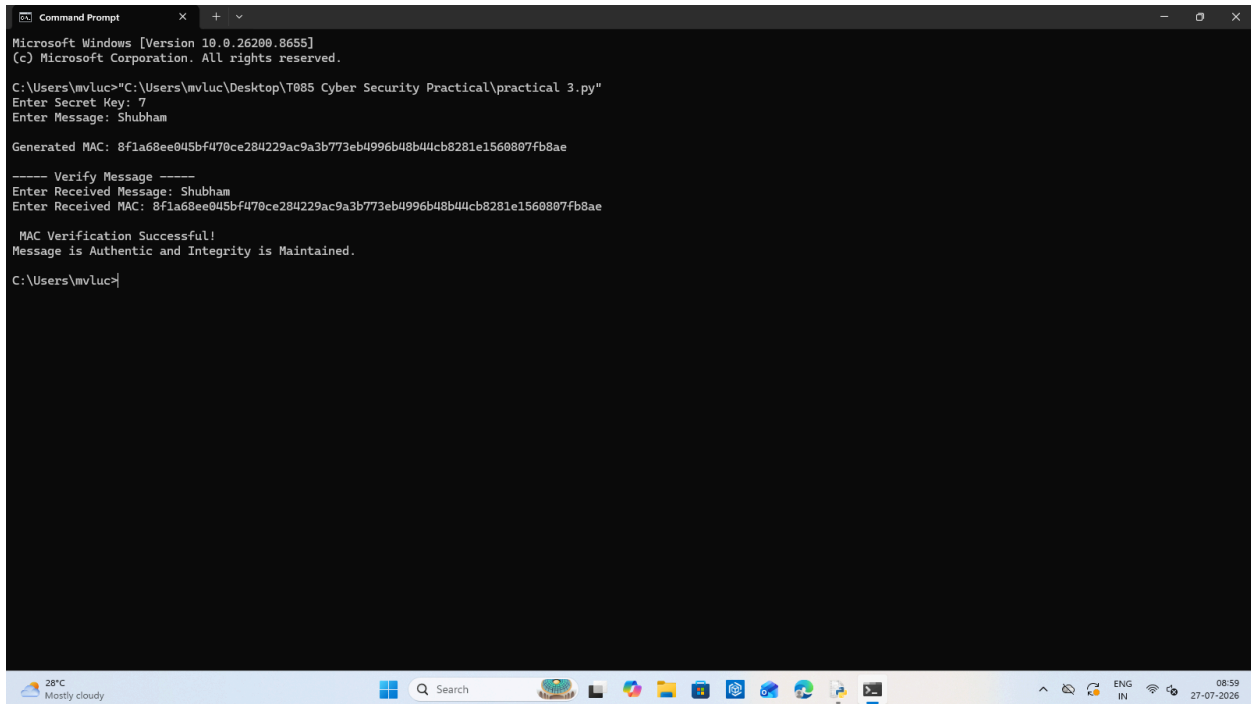
# Verify MAC
if hmac.compare_digest(received_mac, new_mac):
    print("\n MAC Verification Successful!")
    print("Message is Authentic and Integrity is Maintained.")
else:
    print("\n MAC Verification Failed!")
    print("Message has been Modified or Authentication Failed.")
```

Name : Shubham Sanjay Karape

Roll No : T085

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Output :



```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mvluc>"C:\Users\mvluc\Desktop\T085 Cyber Security Practical\practical 3.py"
Enter Secret Key: 7
Enter Message: Shubham

Generated MAC: 8f1a68ee045bf470ce284229ac9a3b773eb4996b48b44cb8281e1560807fb8ae

----- Verify Message -----
Enter Received Message: Shubham
Enter Received MAC: 8f1a68ee045bf470ce284229ac9a3b773eb4996b48b44cb8281e1560807fb8ae

MAC Verification Successful!
Message is Authentic and Integrity is Maintained.

C:\Users\mvluc>
```

GUI Version :

Code :

```
import tkinter as tk
from tkinter import messagebox
import hmac
import hashlib

# ----- Generate MAC -----
def generate_mac():
    key = entry_key.get().encode()
    message = text_message.get("1.0", tk.END).strip().encode()

    if not key or not message:
        messagebox.showerror("Error", "Please enter both Secret Key and Message.")
        return

    mac = hmac.new(key, message, hashlib.sha256).hexdigest()

    entry_generated_mac.config(state="normal")
    entry_generated_mac.delete(0, tk.END)
```

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```
entry_generated_mac.insert(0, mac)
entry_generated_mac.config(state="readonly")
```

```
# ----- Verify MAC -----
def verify_mac():
    key = entry_key.get().encode()
    received_message = text_verify_message.get("1.0", tk.END).strip().encode()
    received_mac = entry_verify_mac.get()

    if not key or not received_message or not received_mac:
        messagebox.showerror("Error", "Please fill all verification fields.")
        return

    new_mac = hmac.new(key, received_message, hashlib.sha256).hexdigest()

    if hmac.compare_digest(received_mac, new_mac):
        result_label.config(
            text="✔ MAC Verification Successful!\nMessage is Authentic.",
            fg="green"
        )
    else:
        result_label.config(
            text="✘ MAC Verification Failed!\nMessage has been Modified.",
            fg="red"
        )
```

```
# ----- GUI -----
root = tk.Tk()
root.title("Message Authentication Code (MAC)")
root.geometry("650x600")
root.configure(bg="black")

title = tk.Label(
    root,
    text="MAC Generator & Verifier",
    font=("Arial", 18, "bold"),
    bg="black",
    fg="cyan"
)
title.pack(pady=10)

# Secret Key
tk.Label(root, text="Secret Key", bg="black", fg="cyan",
        font=("Arial", 11)).pack()
```

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```
entry_key = tk.Entry(root, width=50, bg="#1a1a1a",
                    fg="white", insertbackground="white")
entry_key.pack(pady=5)

# Message
tk.Label(root, text="Message", bg="black", fg="cyan",
        font=("Arial", 11)).pack()

text_message = tk.Text(root, height=4, width=55,
                    bg="#1a1a1a", fg="white",
                    insertbackground="white")
text_message.pack(pady=5)

# Generate Button
tk.Button(root,
        text="Generate MAC",
        command=generate_mac,
        bg="cyan",
        fg="black",
        font=("Arial", 11, "bold")).pack(pady=10)

# Generated MAC
tk.Label(root, text="Generated MAC", bg="black",
        fg="cyan", font=("Arial", 11)).pack()

entry_generated_mac = tk.Entry(root, width=70, state="readonly",
                    bg="#1a1a1a", fg="yellow",
                    readonlybackground="#1a1a1a")
entry_generated_mac.pack(pady=5)

# Verification Section
tk.Label(root,
        text="Verification",
        font=("Arial", 15, "bold"),
        bg="black",
        fg="cyan").pack(pady=15)

tk.Label(root, text="Received Message", bg="black",
        fg="cyan").pack()

text_verify_message = tk.Text(root, height=4, width=55,
                    bg="#1a1a1a",
                    fg="white",
                    insertbackground="white")
text_verify_message.pack(pady=5)

tk.Label(root, text="Received MAC", bg="black",
```

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```
fg="cyan").pack()

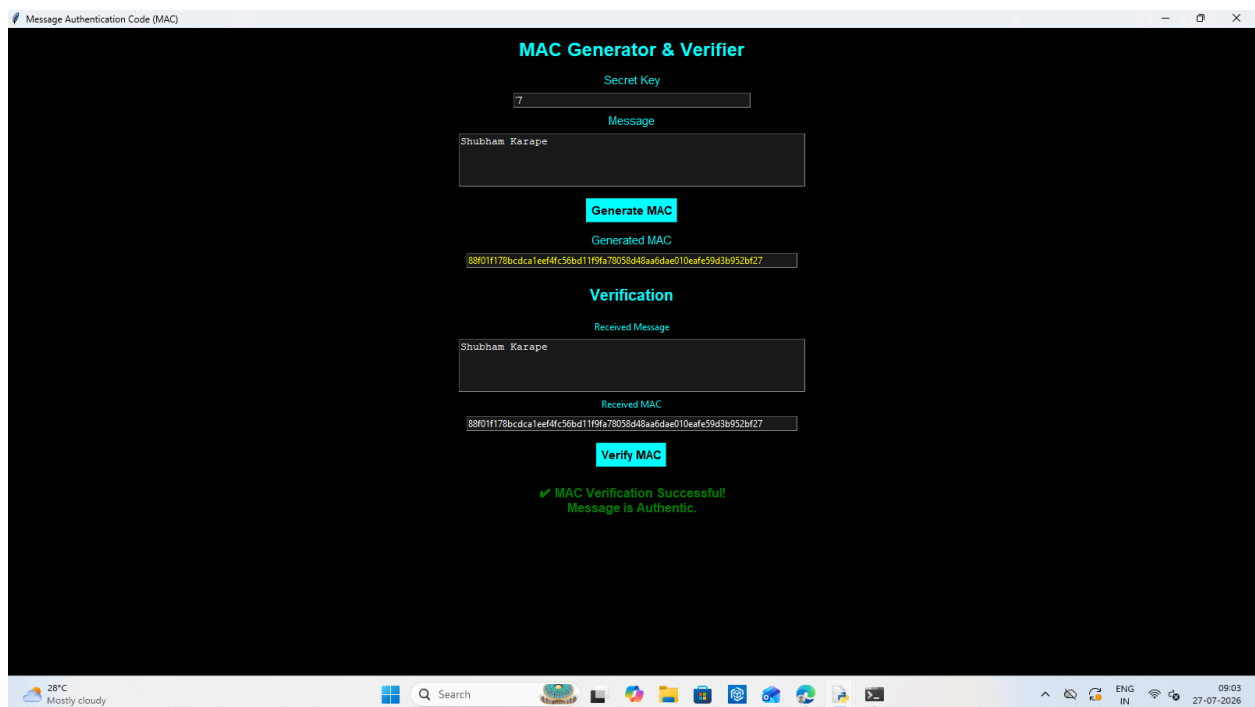
entry_verify_mac = tk.Entry(root, width=70,
                             bg="#1a1a1a",
                             fg="white",
                             insertbackground="white")
entry_verify_mac.pack(pady=5)

# Verify Button
tk.Button(root,
          text="Verify MAC",
          command=verify_mac,
          bg="cyan",
          fg="black",
          font=("Arial", 11, "bold")).pack(pady=10)

# Result Label
result_label = tk.Label(root,
                        text="",
                        font=("Arial", 12, "bold"),
                        bg="black")
result_label.pack(pady=10)

root.mainloop()
```

Output :



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